

Panduan untuk VPN FortiClient IPsec VPN



1. Langkah pertama klik **Garis 3** lalu klik (**add a new connection**)

FortiClient VPN

Upgrade to the full version to access additional features and receive technical support.

New VPN Connection

VPN: SSL VPN **IPsec VPN** Any

Connection Name: DKI IPsecVPN

Description:

Remote Gateway:

+ Add Remote Gateway

Authentication Method: Pre-shared key

Authentication (XAuth): ☒ Prompt on login ☐ Save login ☐ Disable

Fallover SSL VPN: [None]

Single Sign On Settings: ☐ Enable Single Sign On (SSO) for VPN Tunnel

+ Advanced Settings

2. Untuk langkah kedua pilih (**IPsec VPN**) lalu pilih (**Connection Name : DKI IPsecVPN**)

FortiClient VPN

Upgrade to the full version to access additional features and receive technical support.

Edit VPN Connection

VPN: SSL VPN **IPsec VPN** Any

Connection Name: DKI IPsecVPN

Description:

Remote Gateway: 103.209.7.253

+ Add Remote Gateway

Authentication Method: Pre-shared key

Authentication (XAuth): ☒ Prompt on login ☐ Save login ☐ Disable

Fallover SSL VPN: [None]

Single Sign On Settings: ☐ Enable Single Sign On (SSO) for VPN Tunnel

+ Advanced Settings

3. Untuk langkah ketiga isi pada kolom **Remote Gateway** 103.209.7.253 dan isi pada kolom **Pre-Shared Key** : HQ{g}yUA=w@T

FortiClient VPN

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Edit VPN Connection

VPN: SSL VPN IPsec VPN XML

Connection Name:

Description:

Remote Gateway:

Authentication Method: Pre-shared key

Authentication (XAuth): ☒ Prompt on login ☐ Save login ☐ Disable

Fallover SSL VPN: [None]

Single Sign On Settings: ☐ Enable Single Sign On (SSO) for VPN Tunnel

— Advanced Settings

+ VPN Settings

+ Phase 1

+ Phase 2

4. Untuk langkah keempat klik pada + **VPN Settings**

FortiClient VPN

Upgrade to the full version to access additional features and receive technical support.

Edit VPN Connection

VPN: SSL VPN IPsec VPN XML

Connection Name:

Description:

Remote Gateway:

Authentication Method: Pre-shared key

Authentication (XAuth): ☒ Prompt on login ☐ Save login ☐ Disable

Fallover SSL VPN: [None]

Single Sign On Settings: ☐ Enable Single Sign On (SSO) for VPN Tunnel

— Advanced Settings

— VPN Settings

IKE: ☒ Version 1 ☐ Version 2

Mode: ☐ Main ☒ Aggressive

Address Assignment: ☒ Mode Config ☐ Manually ☐ DHCP over IPsec

+ Phase 1

+ Phase 2

5. Untuk langkah selanjutnya pastikan klik seperti panah di atas gambar berikut.

FortiClient VPN

Upgrade to the full version to access additional features and receive technical support.

VPN

SSL-VPN IPsec VPN XML

Connection Name: DKI IPsecVPN

Description:

Remote Gateway: 103.209.7.253

+ Add Remote Gateway

Authentication Method: Pre-shared key

Authentication (XAuth): ☒ Prompt on login ☐ Save login ☐ Disable

Failover SSL VPN: [None]

Single Sign On Settings: ☐ Enable Single Sign On (SSO) for VPN Tunnel

Advanced Settings

VPN Settings

IKE: ☒ Version 1 ☐ Version 2

Mode: ☐ Main ☒ Aggressive

Address Assignment: ☒ Mode Config ☐ Manually Set ☐ DHCP over IPsec

+ Phase 1

+ Phase 2

6. Untuk selanjutnya klik pada + **Phase 1** seperti pada gambar di atas.

FortiClient VPN

Upgrade to the full version to access additional features and receive technical support.

Advanced Settings

VPN Settings

IKE: ☒ Version 1 ☐ Version 2

Mode: ☐ Main ☒ Aggressive

Address Assignment: ☒ Mode Config ☐ Manually Set ☐ DHCP over IPsec

Phase 1

IKE Proposal: Encryption: AES128

Encryption: AES256

DH Group: ☐ 1 ☐ 2 ☐ 5 ☐ 16 ☐ 17 ☐ 18 ☐ 21

Key Life: 86400 sec

Local ID: Optional

☒ Dead Peer Detection

☒ NAT Traversal

☐ Enable Local LAN

+ Phase 2

7. Untuk selanjutnya isi pada kolom pada gambar di atas. **Encryption : AES128** **Encryption AES256** **Authentication SHA256** **Authentication SHA256** dan DH Group isi angka **14** , untuk key life isi angka **86400** , dan **ceklis** pada Dead Peer Detection , dan **ceklis** pada NAT Traversal.



8. Untuk selanjutnya klik pada + **Phase 2** seperti gambar di atas.



9. Untuk langkah selanjutnya ikuti pada panah gambar di atas. **Encryption : AES128** **Encryption: AES256** **Authentication : SHA1** **Authentication : SHA256** , pada Key Life , ceklis isi : **43200** dan di bawah nya isi : **0** , pada enable replay detection untuk di **ceklis** dan enable perfect forward secrecy untuk di **ceklis** dan DH Group isi angka **14**. Lalu untuk selanjutnya klik **Save**.



10. Untuk selanjutnya test login akun vpn pada masing-masing akun user.



Panduan Implementasi IPsec Site-to-Site/Client (StrongSwan) pada Ubuntu 24.x

1. Open terminal
2. Install packet Strongswan dan Charon-systemd
 - apt install strongswan
 - apt install charon-systemd

```
root@elk-2:~# apt install strongswan
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
strongswan is already the newest version (5.9.13-2ubuntu4.24.04.1).
0 upgraded, 0 newly installed, 0 to remove and 111 not upgraded.
root@elk-2:~# apt install charon-
charon-cmd      charon-systemd
root@elk-2:~# apt install charon-systemd
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
charon-systemd is already the newest version (5.9.13-2ubuntu4.24.04.1).
0 upgraded, 0 newly installed, 0 to remove and 111 not upgraded.
root@elk-2:~#
```

3. Buat file baru bernama fgt.conf di /etc/swanctl/conf.d, lalu copy kan script berikut:

```
connections {
    FGT {
        version = 1
        aggressive = yes
        proposals = aes256-sha256-modp2048
        remote_addrs = 103.209.7.253
        vips = 0.0.0.0

        # Ronde autentikasi pertama (Pre-Shared Key / PSK)
        local-1 {
            auth = psk
            id = Username VPN
        }

        # Ronde autentikasi kedua (XAuth)
        local-2 {
            auth = xauth
            id = Username VPN
        }

        remote-1 {
            auth = psk
        }

        children {
            net {
                remote_ts = 0.0.0.0/0
                esp_proposals = aes256-sha256-modp2048, aes256-sha1-modp1024, aes256-sha256
                dpd_action = restart
                start_action = start
            }
        }
    }
}

secrets {
    # PSK untuk Fase 1
    ike-fgt {
        id-1 = 103.209.7.253
        secret = "Hq{g}UA=w@T"
    }

    # Password untuk XAuth (User-Based)
    xauth-fgt {
        id-1 = Username VPN
        secret = "Password VPN"
    }
}
```

Note : pengisian credential user dan pass
word vpn didalam script diatas



```
0 upgraded, 0 newly installed, 0 to remove and 111 not upgraded.
root@elk-2:~# cd /etc/swanctl
root@elk-2:/etc/swanctl# cd conf.d
root@elk-2:/etc/swanctl/conf.d# ls
root@elk-2:/etc/swanctl/conf.d# nano fgt.conf
root@elk-2:/etc/swanctl/conf.d#
```

4. Lalu restart strongswan dan start menggunakan perintah berikut

- systemctl start strongswan
- swanctl --load-all
- swanctl --initiate --child net

```
successfully loaded 1 connections, 0 unloaded
root@elk-2:~# systemctl start strongswan
root@elk-2:~# swanctl --load-all
plugin 'test-vectors': failed to load - test_vectors_plugin_create not found and no plugin file available
plugin 'ldap': failed to load - ldap_plugin_create not found and no plugin file available
plugin 'pkcs11': failed to load - pkcs11_plugin_create not found and no plugin file available
plugin 'rdrand': failed to load - rdrand_plugin_create not found and no plugin file available
plugin 'gcrypt': failed to load - gcrypt_plugin_create not found and no plugin file available
plugin 'af-alg': failed to load - af_alg_plugin_create not found and no plugin file available
plugin 'curve25519': failed to load - curve25519_plugin_create not found and no plugin file available
plugin 'chapoly': failed to load - chapoly_plugin_create not found and no plugin file available
plugin 'cmac': failed to load - cmac_plugin_create not found and no plugin file available
plugin 'ctr': failed to load - ctr_plugin_create not found and no plugin file available
plugin 'ccm': failed to load - ccm_plugin_create not found and no plugin file available
plugin 'ntru': failed to load - ntru_plugin_create not found and no plugin file available
plugin 'curl': failed to load - curl_plugin_create not found and no plugin file available
loaded ike secret 'ike-fgt'
loaded xauth secret 'xauth-fgt'
no authorities found, 0 unloaded
no pools found, 0 unloaded
loaded connection 'FGT'
successfully loaded 1 connections, 0 unloaded
root@elk-2:~# _
```

```
successfully loaded 1 connections, 0 unloaded
root@elk-2:~# swanctl --initiate --child net
plugin 'test-vectors': failed to load - test_vectors_plugin_create not found and no plugin file available
plugin 'ldap': failed to load - ldap_plugin_create not found and no plugin file available
plugin 'pkcs11': failed to load - pkcs11_plugin_create not found and no plugin file available
plugin 'rdrand': failed to load - rdrand_plugin_create not found and no plugin file available
plugin 'gcrypt': failed to load - gcrypt_plugin_create not found and no plugin file available
plugin 'af-alg': failed to load - af_alg_plugin_create not found and no plugin file available
plugin 'curve25519': failed to load - curve25519_plugin_create not found and no plugin file available
plugin 'chapoly': failed to load - chapoly_plugin_create not found and no plugin file available
plugin 'cmac': failed to load - cmac_plugin_create not found and no plugin file available
plugin 'ctr': failed to load - ctr_plugin_create not found and no plugin file available
plugin 'ccm': failed to load - ccm_plugin_create not found and no plugin file available
plugin 'ntru': failed to load - ntru_plugin_create not found and no plugin file available
plugin 'curl': failed to load - curl_plugin_create not found and no plugin file available
[ENC] generating QUICK_MODE request 1321202500 [ HASH SA No KE ID ID ]
[NET] sending packet: from 192.168.253.132[4500] to 103.209.7.253[4500] (460 bytes)
[NET] received packet: from 103.209.7.253[4500] to 192.168.253.132[4500] (444 bytes)
[ENC] parsed QUICK_MODE response 1321202500 [ HASH SA No KE ID ID ]
[CFG] selected proposal: ESP_AES_CBC_256/HMAC_SHA2_256_128/MODP_2048/NO_EXT_SEQ
[IKE] detected rekeying of CHILD_SA net{1}
[IKE] CHILD_SA net{2} established with SPIs c42a49c4_i 1902f214_o and TS 172.16.0.0/32 === 0.0.0.0/0
Initiate completed successfully
root@elk-2:~#
```

5. Untuk memutus koneksi dapat menjalankan perintah berikut

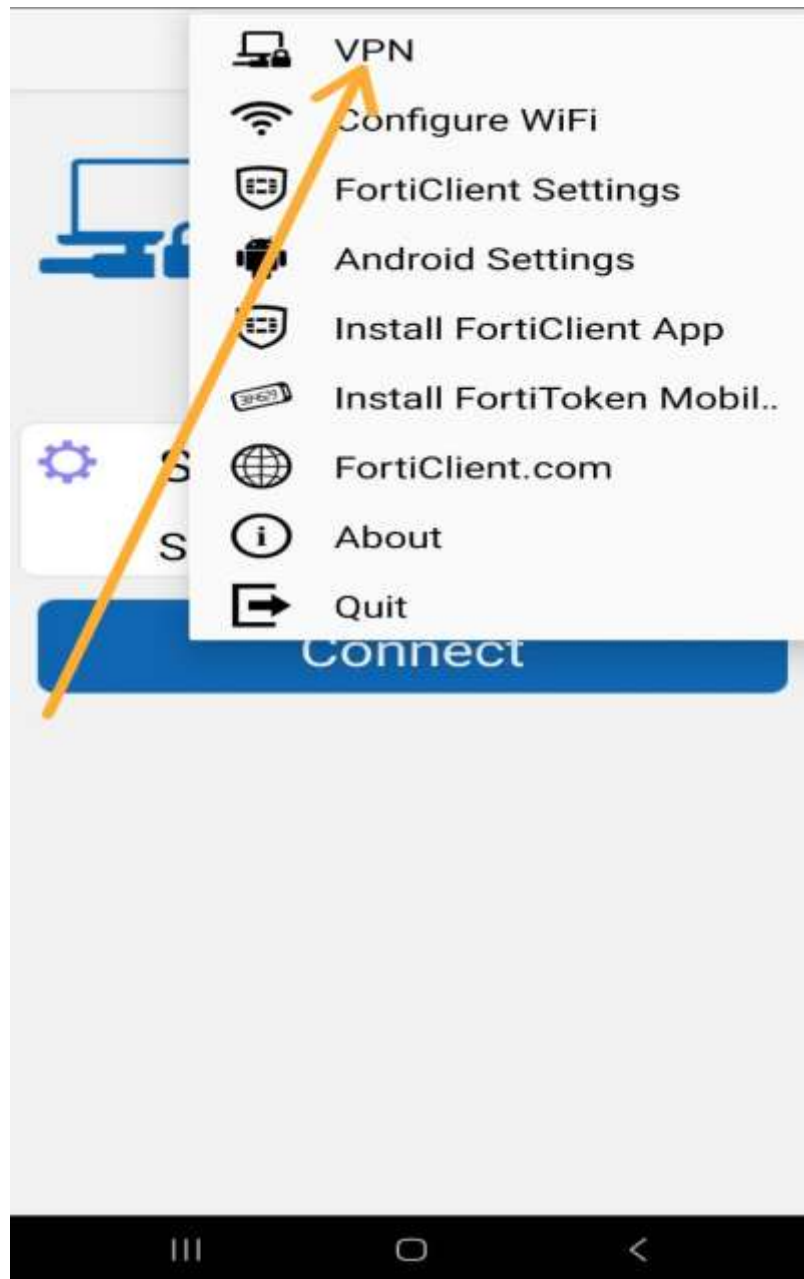
Swanctl --terminate --child net

```
root@elk-2:~# swanctl --terminate --child net
plugin 'test-vectors': failed to load - test_vectors_plugin_create not found and no plugin file available
plugin 'ldap': failed to load - ldap_plugin_create not found and no plugin file available
plugin 'pkcs11': failed to load - pkcs11_plugin_create not found and no plugin file available
plugin 'rdrand': failed to load - rdrand_plugin_create not found and no plugin file available
plugin 'gcrypt': failed to load - gcrypt_plugin_create not found and no plugin file available
plugin 'af-alg': failed to load - af_alg_plugin_create not found and no plugin file available
plugin 'curve25519': failed to load - curve25519_plugin_create not found and no plugin file available
plugin 'chapoly': failed to load - chapoly_plugin_create not found and no plugin file available
plugin 'cmac': failed to load - cmac_plugin_create not found and no plugin file available
plugin 'ctr': failed to load - ctr_plugin_create not found and no plugin file available
plugin 'ccm': failed to load - ccm_plugin_create not found and no plugin file available
plugin 'ntru': failed to load - ntru_plugin_create not found and no plugin file available
plugin 'curl': failed to load - curl_plugin_create not found and no plugin file available
[IKE] closing CHILD_SA net{1} with SPIs c9813541_i (76 bytes) 1902f20c_o (76 bytes) and TS 172.16.0.0/32 === 0.0.0.0/0
[IKE] closing CHILD_SA net{2} with SPIs c42a49c4_i (0 bytes) 1902f214_o (0 bytes) and TS 172.16.0.0/32 === 0.0.0.0/0
terminate completed successfully
root@elk-2:~#
```

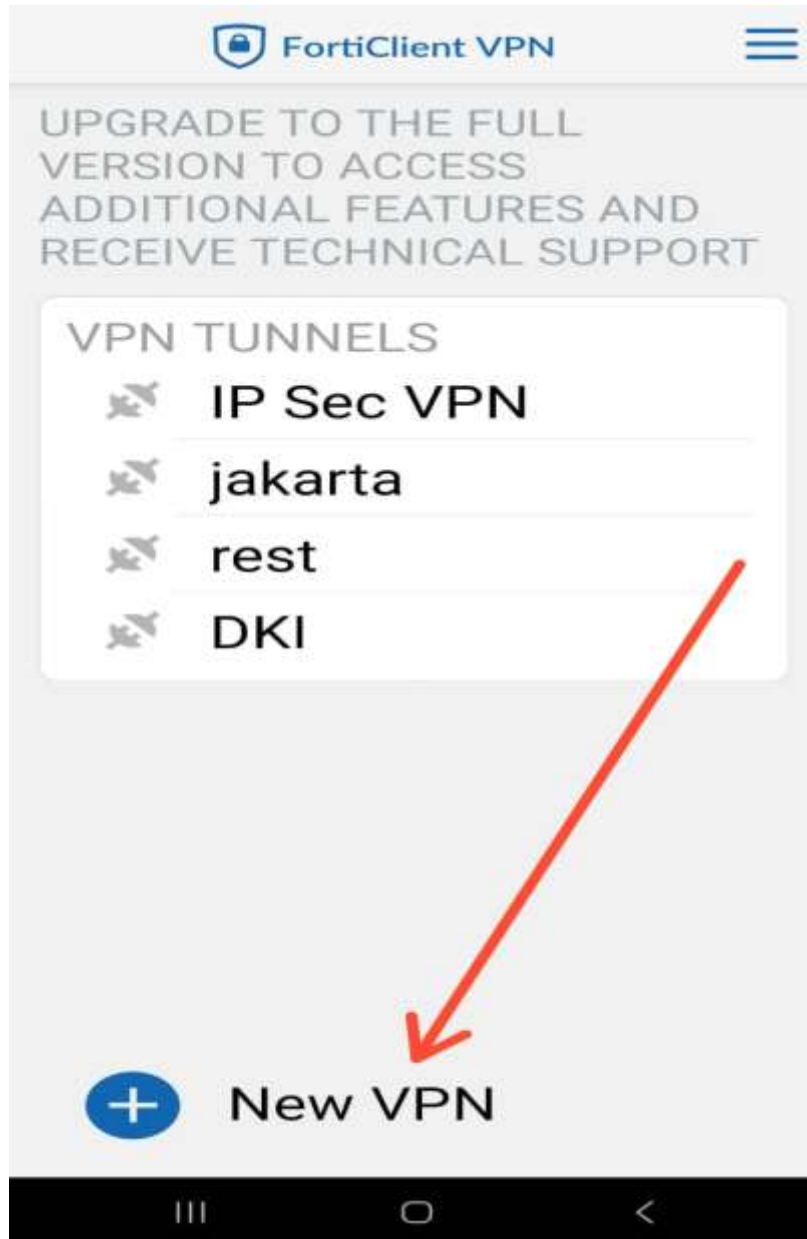
Panduan untuk VPN FortiClient IPsec VPN Android



1. Langkah pertama klik **Garis 3** lalu klik.



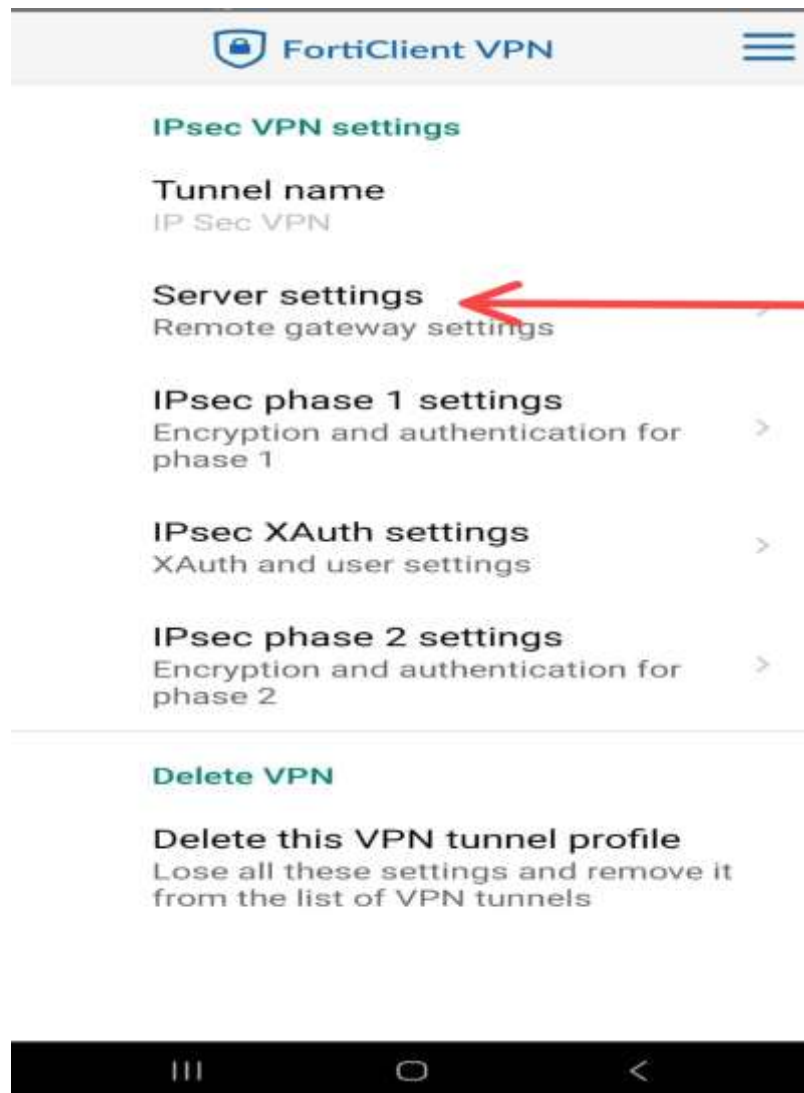
2. Untuk langkah kedua pilih **(VPN)**.



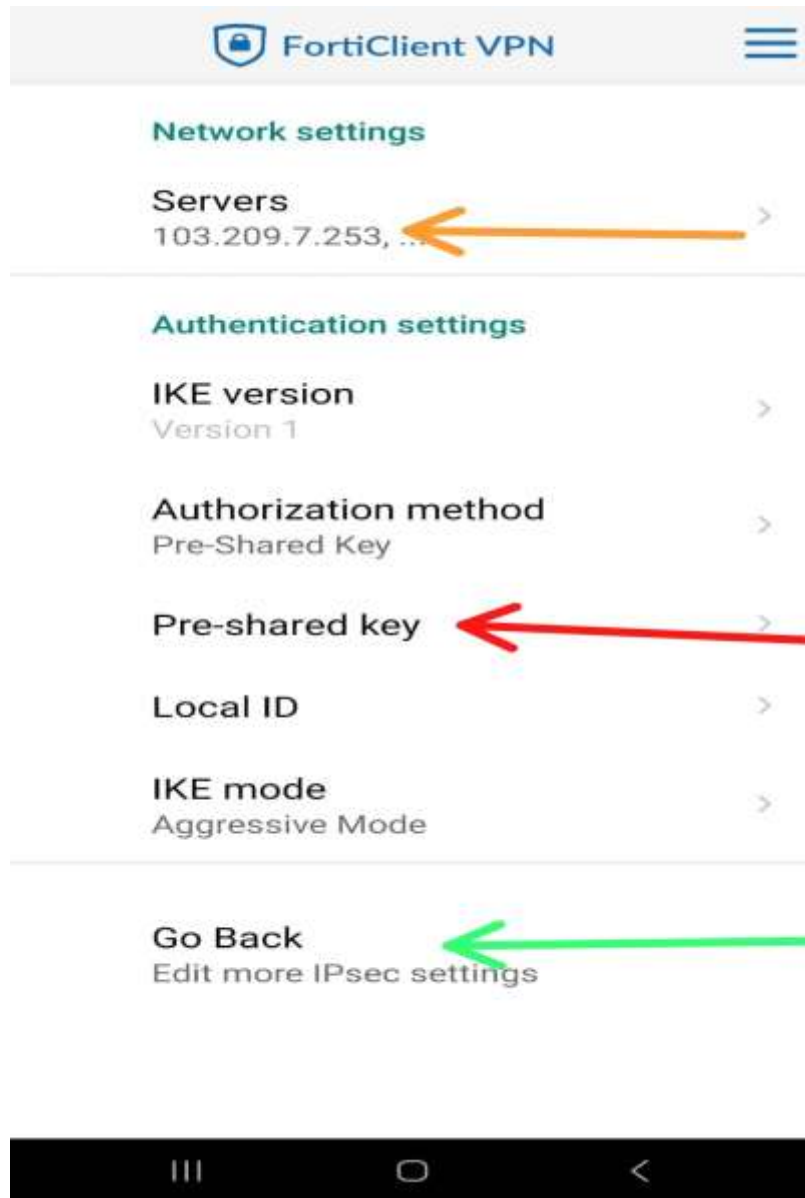
3. Untuk langkah ketiga pilih (**+ New VPN**).



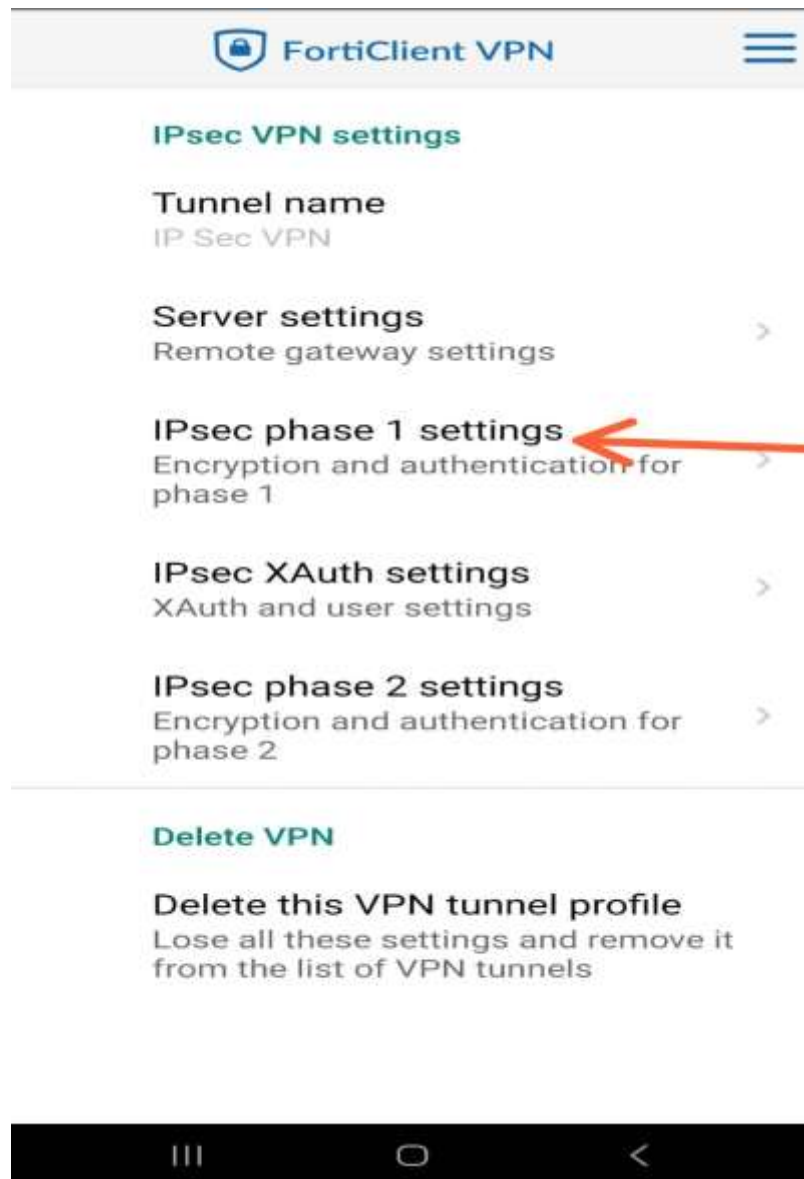
4. Untuk langkah keempat isi **Add VPN** dengan nama (**IP Sec VPN**) , Lalu pilih **VPN Type** (**IPsec VPN**) Setelah itu **klik Create**.



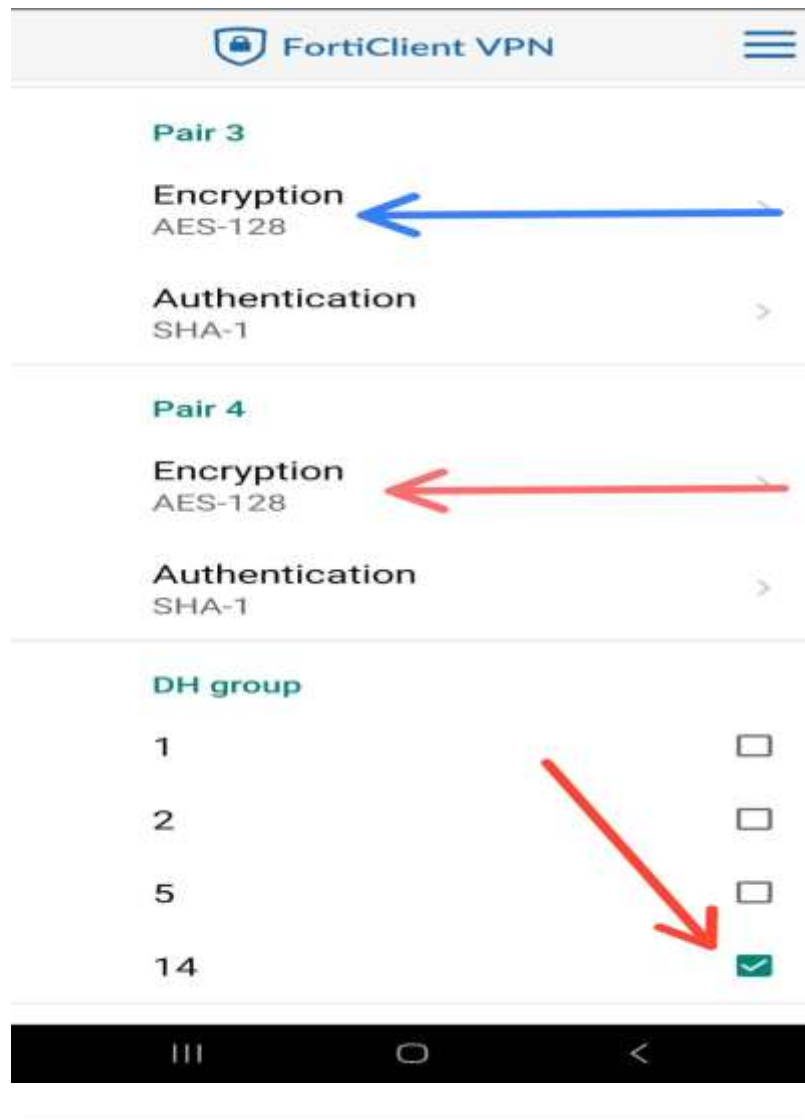
5. Untuk langkah kelima pilih **Server setting**



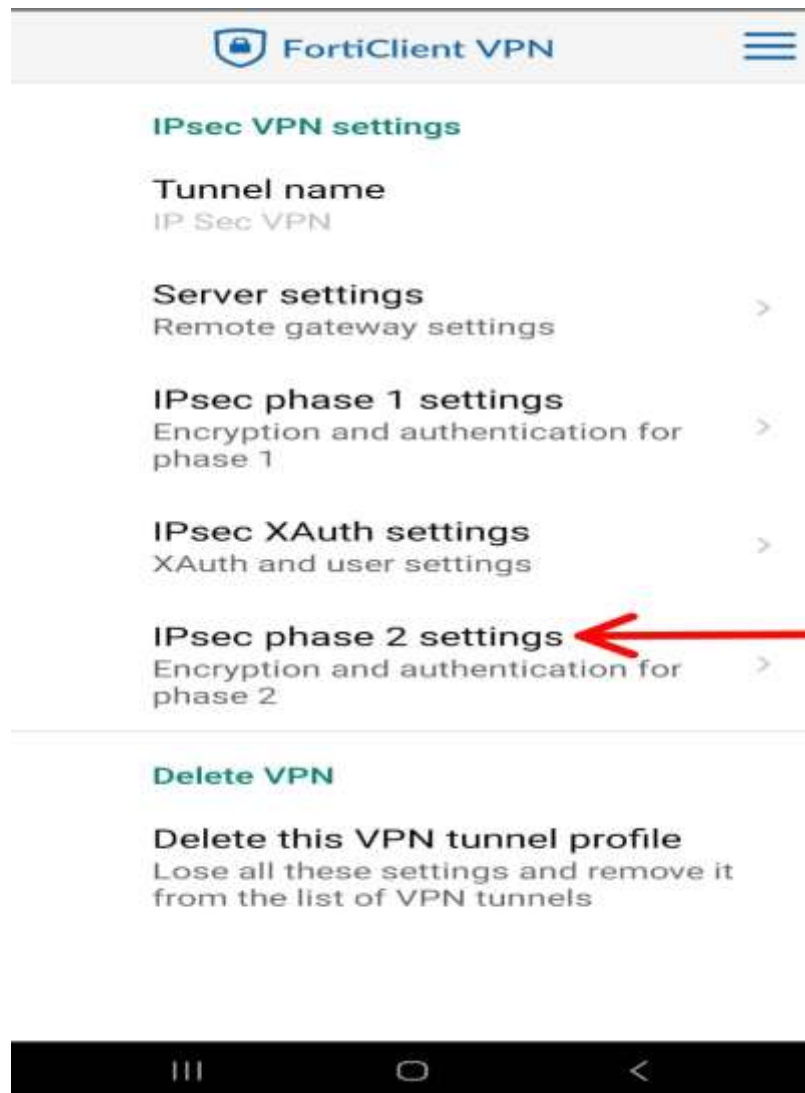
6. Untuk langkah keenam isi **Servers (103.209.7.253)**. Lalu Pilih **Pre-shared key** isi dengan : **HQ{g]yUA=w@T** ,selanjutnya pilih **Go Back**.



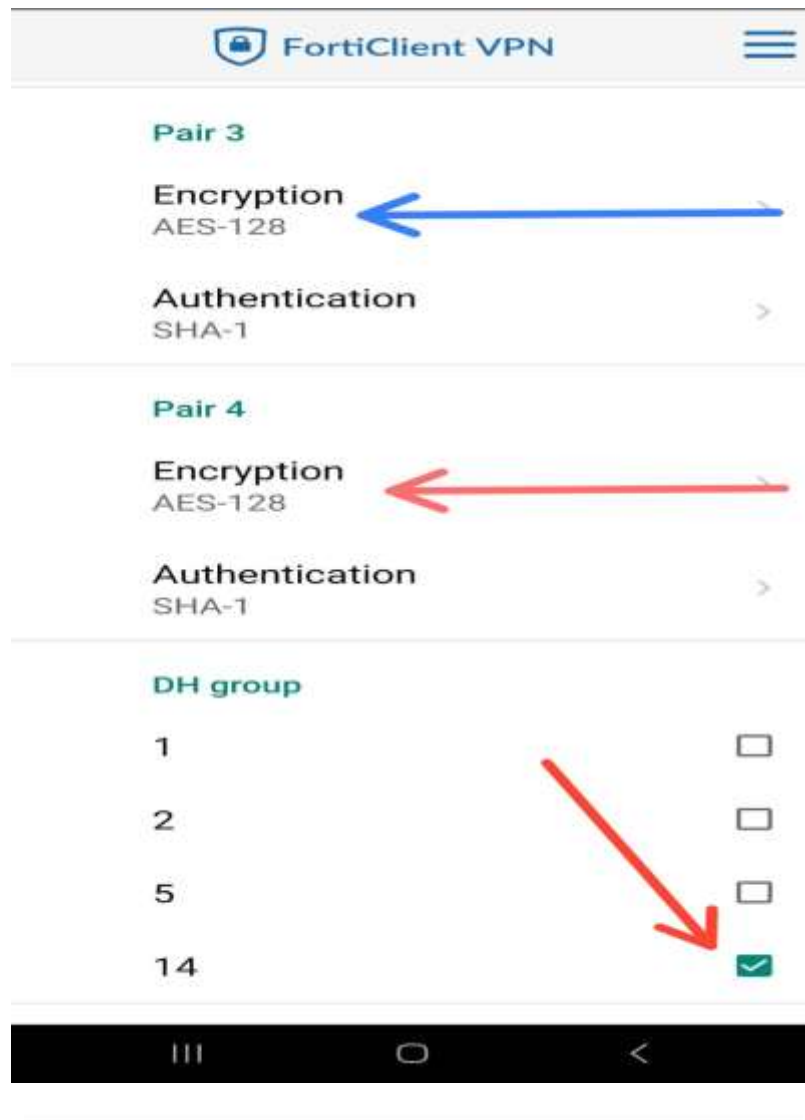
7. Untuk langkah ketujuh pilih **IPsec phase 1 settings**.



8. Untuk langkah kedelapan pilih **Pair 3** lalu pilih untuk **Encryption** pilih **AES-128** dan untuk **Pair 4** pilih untuk **Encryption** pilih **AES-128** setelah itu untuk **DH group** ceklis angka **14**. Lalu pilih **Go Back**



9. Untuk langkah kesembilan pilih **IPsec phase 2 settings**.



10. Untuk langkah kesepuluh pilih **Pair 3** lalu pilih untuk **Encryption** pilih **AES-128** dan untuk **Pair 4** pilih untuk **Encryption** pilih **AES-128** setelah itu untuk **DH group** ceklis angka **14**. Lalu pilih **Go Back**. Setelah itu test login akun vpn pada masing-masing akun user.



11. test VPN IPsec pada masing-masing akun user.